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A Universal, EOS-Based Correlation for Z-Factor, Viscosity and Enthalpy for Hydrocarbon and H₂, N₂, CO₂, H₂S Gas Mixtures

M. W. Burgoyne, Santos Ltd, Australia; M. H. Nielsen and M. Stanko, Whitson AS, Norway

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Abstract

Accurate estimation of gas compressibility (Z-Factor), viscosity, and enthalpy is vital across conventional hydrocarbon production, greenhouse gas storage, and hydrogen energy applications. Legacy Z-factor correlations like Dranchuk-Abou-Kassem (DAK) and Hall-Yarborough (HY) perform well for typical natural gases but lose accuracy - and sometimes fail - at low temperatures, high pressures, or with inert-rich mixtures. Advanced frameworks such as GERG2008 address a broader range of compositions but are often impractical for spreadsheets and standard engineering workflows due to software, complexity, and compositional constraints.

We present a unified thermodynamic model combining a volume-translated Peng-Robinson Equation of State (EOS) for Z-factor, density, enthalpy, Cp, and Joule-Thomson coefficient with the Lohrenz-Bray-Clark (LBC) method for viscosity. The model covers one hydrocarbon and four inert components (H₂, N₂, CO₂, H₂S), with parameters tuned to Standing-Katz, NIST, binary vapor liquid equilibrium (VLE) and publicly available experimental Z-Factor data. Validation shows accuracy equal to or better than traditional correlations from 50-300 °F and 14.7-15,000 psia, including gas condensates and associated hydrocarbon gases as well as inert and hydrogen-rich mixtures. Viscosity results closely match reference data.

This approach streamlines gas property calculations with a single, analytically solved, spreadsheet-ready framework - facilitating integration into reservoir simulators and meeting the needs of both conventional and emerging carbon dioxide and hydrogen storage workflows.

Additionally, in its simplest form, this analytically solvable cubic EOS-based method can be a compact, direct and robust replacement for iterative legacy methods (e.g., DAK, HY), eliminating numerical convergence issues and simplifying integration into standard engineering workflows.

All calculations, regression data, and example implementations in Python and Excel VBA are available on GitHub.

Introduction

While legacy Z-factor correlations such as DAK ([Dranchuk and Abou-Kassem 1975](#)) and HY ([Hall 1973](#)) remain standard for sweet natural gases, they exhibit significant limitations when applied to mixtures with high fractions of CO₂, N₂, H₂S, or H₂. At low reduced temperatures or high inert content, these models may

become inaccurate or fail to converge, and they lack direct support for hydrogen-rich systems. Likewise, conventional viscosity models - including Lee-Gonzalez-Eakin (Lee A.L. 1966) and LBC (Lohrenz J. 1964) - are reliable only for light, hydrocarbon-dominated gases, while thermodynamic properties such as enthalpy and specific heat are seldom addressed in spreadsheet-based solutions.

Advanced EOS frameworks like GERG2008 (Kunz 2007) and pure-fluid models such as Span-Wagner (Span 1996) provide comprehensive coverage and high accuracy for Z-factor, viscosity, and thermal properties, but their deployment in routine engineering remains limited by software constraints, complexity, and the need for detailed compositional analysis. In practice, this forces engineers to combine several legacy methods and ad hoc corrections when working with unconventional gas mixtures or emerging carbon capture and underground storage (CCUS) and hydrogen projects.

To bridge this gap, the present model integrates a volume-translated Peng-Robinson EOS for Z-factor and thermal properties with a tuned Lohrenz-Bray-Clark viscosity model, all using a single set of physically meaningful parameters. Model constants - including temperature-dependent BIPs - are regressed to Standing-Katz, National Institute of Standards and Technology (NIST), and VLE data for a five-component system (hydrocarbon pseudocomponent, H₂, N₂, CO₂, H₂S). The result is a practical, spreadsheet-compatible tool that delivers robust predictions for both conventional and inert-rich or hydrogen-containing gases.

Analytical solution of the cubic EOS enables rapid single-phase property estimation suitable for spreadsheets and scripting environments, while rigorous flash calculations are supported for phase equilibrium scenarios using the same tuned parameters. High accuracy is maintained for single-phase predictions over an extensive range of conditions and compositions. For complex phase behavior in heavy hydrocarbon systems, more specialized EOS models would still be preferred.

In its simplest single component form, the proposed analytically solved PR-EOS approach also offers a practically advantageous alternative to iterative legacy correlations, as demonstrated explicitly in Appendix B. This simplification yields immediate benefits in spreadsheet-based or rapid-calculation workflows.

Literature Review

Empirical Z-Factor Correlations: Historical Development and Numerical Issues

The estimation of natural gas compressibility factors (Z) has been central to gas engineering since the publication of the generalized Z-factor chart by Standing and Katz (Standing 1942), which became the foundation for routine field and laboratory calculations for decades. Recognizing the need for easier digital calculation, Hall and Yarborough (Hall 1973) introduced an analytic functional form fitted to the Standing-Katz chart, and Dranchuk and Abou-Kassem (Dranchuk and Abou-Kassem 1975) advanced the field with an 11-parameter implicit equation, enabling rapid and (in principle) robust calculation of Z for a wide range of pseudo-reduced temperatures and pressures.

Since DAK, numerous alternative empirical Z-factor correlations have been published - each claiming to improve accuracy or extend the range of conditions - but most fundamentally reproduce the behavior of the Standing-Katz data, and few demonstrably outperform a well-implemented DAK method. In practice, DAK remains a benchmark for natural gas Z-factor prediction due to its flexibility and reliability - provided the iterative numerical solution is properly constrained.

A recurring theme in the literature e.g., (Mahmoud 2013) is that DAK and similar implicit correlations "fail" or produce nonphysical results at high pressures, with reported errors sometimes exceeding 100% above 15,000 psia. However, these failures are almost always artifacts of the numerical solution method - specifically, poor initial guesses or insufficient convergence controls in the iterative Z-solver - rather than any fundamental defect in the DAK solution itself. Our experience shows that, with robust initialization and convergence criteria, DAK delivers accurate Z-factors even at very high pressures.

The true numerical challenges with DAK and related methods emerge at low reduced temperatures ($T_r < 1$), where the equation's strong nonlinearity requires especially careful initialization and constraints for successful convergence. This underscores a practical limitation of all iterative implicit Z-factor correlations.

The present work advances beyond these legacy methods by employing an analytic solution of the cubic Peng-Robinson EOS for Z, avoiding iterative schemes and guaranteeing stable and accurate results across all single-phase conditions, regardless of pressure or temperature.

Critical Property Mixing and Pseudocomponent Methods

Accurate Z-factor estimation for gas mixtures depends on reliable calculation of mixture pseudocritical temperature (T_c) and pressure (P_c), which are then used to compute pseudo-reduced properties. The most widely used approach, Kay's mixing rule (Kay 1936), averages the pure component critical properties on a molar basis. While Kay's method is effective when full compositional data are available, it is less practical when only global parameters such as gas molecular weight (MW) or specific gravity (SG) and inerts content are known.

To address this limitation, numerous empirical correlations have been proposed to estimate pseudocritical properties directly from MW or SG - most notably by Sutton (R. Sutton 1985) and (R. Sutton 2007) and Piper, McCain, Holditch & Corredor (Piper, et al. 1993). These methods enable engineers to estimate T_c and P_c using readily available field data, making them suitable for rapid or scoping calculations where detailed composition is not reported.

A fundamental limitation of all such approaches - from Kay's to Sutton and PMC - is that they rely on averaging the diverse critical properties of the mixture's components to a single T_c and P_c value to represent the mixture. This averaging process introduces challenges, particularly for mixtures containing components with significantly different critical properties, such as the light inert gases (CO_2 , H_2S , N_2 , H_2) versus hydrocarbons. Furthermore, even within the hydrocarbon fraction, variability in hydrocarbon origin (gas condensate rich in LPGs vs associated gas rich in C_7^+) can impact the representative critical properties, adding a layer of uncertainty that is not resolvable when working solely from mixture or hydrocarbon MW.

An often-overlooked limitation of many empirical correlations is that they do not yield the critical properties of pure methane when evaluated with zero inerts at methane's molecular weight. This "lack of methane fallback" can lead to small but systematic bias in Z- factor and property estimation for dry or nearly pure methane gases. Although this behavior is not noted in the original publications, it is evident from analytical evaluation of the correlation equations.

For mixtures containing significant amounts of CO_2 and H_2S (acid gases), further complications arise. The Wichert & Aziz (E. a. Wichert 1971) correction remains widely used to adjust pseudocritical properties for acid-gas-rich mixtures, applying empirical correction factors to Kay's or other base rules. However, as Elsharkawy (2001, 2004) demonstrated, even with such corrections, traditional mixing rules -including Kay's and Sutton's - often underestimate T_c and overestimate P_c in sour or acid-gas-rich systems. This results in incorrect pseudo-reduced properties, even when the Z-factor correlation itself is accurate for a given T_r and P_r .

The approach used in this work substantially addresses these issues by explicitly representing the major inerts (CO_2 , H_2S , N_2 , H_2) and the hydrocarbon pseudocomponent within the Peng-Robinson EOS formulation, leveraging robust Peng Robinson mixing rules that incorporate acentric factors and binary interaction coefficients, and introducing new hydrocarbon critical parameter estimation functions depending upon gas origin (gas condensate or associated gas). This addresses the specific challenges associated with averaging highly disparate critical properties, while still retaining practical implementation for cases where only MW are known.

The new critical property correlations for hydrocarbon gases build on Sutton's insight that associated gases and gas condensates exhibit distinct critical property trends. Our approach employs a monotonic functional form for T_c to ensure physical consistency, and a P_c relationship that capitalizes on the

known near-linear dependence of molar volume on molecular weight. This framework offers more robust extrapolation to high molecular weights, while guaranteeing an exact "methane fallback" at MW = 16.042g/mol.

Advanced EOS Models

While empirical Z-factor correlations are widely used for gas property estimation, cubic equation of state (EOS) models such as the Soave-Redlich-Kwong (SRK) (Soave 1972) and Peng-Robinson (PR) (Peng 1976) equations are standard in petroleum and chemical engineering for rigorous thermodynamic predictions. These models are valued for their mathematical simplicity and ability to predict vapor-liquid equilibria, densities, and compressibility factors for a wide range of fluids.

For highly nonideal mixtures, systems containing significant inerts or polar components, multiparameter EOS models such as GERG- 2008 (Kunz 2007) or the Span-Wagner EOS for CO₂ (Span 1996) can provide higher accuracy. These models require extensive experimental data and are typically used in specialized software tools such as NIST REFPROP (E. B. Lemmon 2018).

In practice, the application of advanced EOS models outside commercial or laboratory software may be limited by availability within the Petroleum Engineer's commercial toolset.

Viscosity Models

Classical correlations for gas-phase viscosity in petroleum engineering include the Carr, Kobayashi, and Burrows charts (Carr 1954) for light hydrocarbons and the Lee, Gonzalez, and Eakin equation (Lee A.L. 1966), which requires only gas specific gravity, z-factor and temperature. These methods provide reasonable estimates for sweet natural gases at low to moderate pressures.

The Lohrenz, Bray, and Clark (Lohrenz J. 1964) model introduced a compositional approach based on the principle of corresponding states, using the critical properties and mole fractions of each component. LBC remains widely implemented in compositional reservoir simulators and is applicable to a broad range of mixtures. However, its predictive accuracy is sensitive to the quality of component property data and to the empirical coefficients used, and it was originally validated against a limited set of experimental data.

Modern reference data, such as those provided by NIST (National Institute of Standards and Technology n.d.), employ specialized viscosity correlations for each pure component (see Table 1), providing highly accurate predictions for methane, hydrogen, nitrogen, carbon dioxide, and hydrogen sulphide across a wide range of temperatures and pressures.

Table 1—NIST Viscosity models for pure components of interest in proposed model

Pure Component	NIST WebBook Viscosity Model
CH ₄	(Quinones-Cisneros 2012)
H ₂	(Muzny 2013)
N ₂	(E. a. Lemmon 2004)
CO ₂	(Laesecke and Muzny 2017)
H ₂ S	(Schmidt, et al. 2008)

For mixtures, the NIST REFPROP package (E. B. Lemmon 2018) employs an Extended Corresponding States (ECS) viscosity model (Huber 2018), which leverages high-quality pure component data and mixture theory to provide reliable predictions. This model, however, is not generally available in most subsurface engineering or spreadsheet-based tools.

CoolProp (Bell 2014), an open-source thermophysical property library, implements the Chung (Chung 1988) model for dilute gas viscosity and applies mixing rules to estimate mixture viscosity. This approach is suitable for engineering calculations but is also less widely available in industry-standard compositional simulators.

This work has found that default LBC viscosity predictions can substantially deviate from NIST reference data, especially outside the envelope of the original LBC dataset. However, with suitable tuning of the LBC model parameters, accuracy can be substantially improved for a wide range of hydrocarbon and non-hydrocarbon mixtures.

Thermal Properties

Scoping level engineering workflows often used fixed or simplified values for enthalpy, isobaric heat capacity (Cp), and the Joule-Thomson (JT) coefficient, which generally sufficed for primary production of natural gas systems. For more accurate property estimation, departure function methods using cubic equations of state (EOS) are available in thermodynamic texts (Prausnitz 1998), (Reid 2000), enabling calculation of enthalpy and Cp for real gases.

Ideal gas heat capacities for pure hydrocarbons are typically represented by polynomial correlations (Riazi 2005). However, published Cp correlations for gas mixtures are less common, and those available are often limited in compositional scope, particularly for inerts and hydrogen. Advanced EOS frameworks such as GERG2008 and Span-Wagner provide high-fidelity thermal properties for specific gases but require full compositional analysis and are rarely accessible for routine calculations in spreadsheet environments.

There remains a practical gap in the availability of models or correlations designed for straightforward implementation across the diverse range of hydrocarbon, inert, and hydrogen-rich mixtures encountered in modern gas production and storage workflows.

Methodology and Model Development

Overview of the proposed Model

The foundation is a Peng-Robinson (PR) equation of state enhanced with volume translation to improve density predictions. We specifically leverage the PR EOS's analytic nature: for a given gas composition, temperature, and pressure, the cubic EOS can be solved explicitly for the largest real root without iteration. By always selecting the largest root (assuming single-phase conditions), we avoid complex flash calculations (no Rachford-Rice or fugacity iterations needed). This makes the implementation straightforward in Excel or programming libraries. Volume translation entails adding a shift to the molar volume for each component, allowing improved density matching while binary interaction coefficients ensure that mixtures of pure methane, CO₂, H₂S, N₂ and H₂ exhibit experimentally consistent VLE behaviour when implemented in a full PVT model. The proposed model supports five components: one hydrocarbon pseudocomponent representing the lumped C_i⁺ hydrocarbon fraction, plus inerts H₂, N₂, CO₂ and H₂S.

While this model is robust for mixtures predominantly composed of methane, inert gases, and moderate fractions of heavier hydrocarbons, caution is recommended for mixtures with significant proportions of heavier hydrocarbons near dew points or retrograde regions. In such scenarios, dedicated compositional PVT software with explicit flash calculations should be considered for accurate predictions of phase behavior and densities in two phase regions.

Calibration Data and Strategy

The calibration of the EOS model was conducted in six sequential steps, each building incrementally upon previous parameterizations:

1. Pure Hydrocarbon Gases: Establish baseline EOS parameters for a single hydrocarbon pseudocomponent.
2. Pure Inerts: Adjust critical parameters specifically for inert gases (H₂, N₂, CO₂, H₂S).
3. Binary Interaction Parameters (BIPs): Refine BIPs using VLE data for binary and ternary mixtures.

4. New Hydrocarbon Pseudocomponent Correlations: Developing critical property correlations for hydrocarbon pseudocomponents of (a) gas condensates and (b) associated gas liberated from oil as a function of MW.
5. Viscosity Tuning: Adjust LBC viscosity model parameters to match NIST data and low pressure Lee Gonzalez & Eakin predictions.
6. Ideal gas heat capacity tuning to NIST data

Steps 1 - 5 were conducted using a flexible commercial PVT tool (Zick, PhazeComp 2012) that permits variable data point weighting and command line driven workflows.

For each calibration step, different parameters were regressed and different weightings used as outlined individually below. For all steps however, the general form of the objective function was as implemented in the commercial PVT tool used, using the following approach (Zick, Residual Weighting Within PhazeComp 2007).

Objective Function to be minimized Reported root-mean-square (RMS) residual Residuals calculated as

$$\chi^2 = \sum_{i=1}^N (w_i r_i)^2 \qquad r = \sqrt{\frac{\chi^2}{\sum_{i=1}^N w_i^2}} \qquad r_i = 100(y_i^c - y_i^e)/y_i^r$$

Where y_i^e is the target value, y_i^c is the calculated value, and y_i^r is the reference value for observation "i". For an isolated measurement, such as the saturation pressure, the reference value will be the same as the experimental value. For a collection of related measurements, however, such as those within the same column of an experiment's data table, the reference value will be taken as the experimental value with the largest magnitude from among that collection of related measurements.

The weight factor (w_i) for an isolated measurement is normally specified with the measurement. For a measurement within an experiment's data table, however, the weight factor is taken as the product of the weight factor assigned to the measurement's column and the weight factor assigned to its row. All weight factors default to 1 unless the user specifies otherwise.

Data decks used, complete with their weightings, can be found on the GitHub site ([Burgoyne, 5_Component_PengRobinson_Z-Factor 2025](#)), from which interested parties can reproduce these fits and/or test impact of alternative weightings.

Step 1: Pure Hydrocarbon gases. We digitized the 5,940 data points from Standing and Katz Z-factor chart data from Table A-2 of Handbook of Natural Gas Engineering ([Katz 1959](#)) - spanning reduced temperatures (Tr) from 1.05 to 3.0 and reduced pressures (Pr) from 0.2 to 15.0. This classic dataset, defined exclusively in terms of reduced properties, serves as the primary reference for pure hydrocarbon natural gas mixtures.

To fit the hydrocarbon pseudocomponent for use with the Peng-Robinson (PR) equation of state, we defined an *arbitrary reference substance* with critical temperature $T_c = 1$ °R and critical pressure $P_c = 1$ psia. This approach ensures that, for any point on the Standing-Katz chart, the reduced temperature ($Tr = T/T_c$) and reduced pressure ($Pr = P/P_c$) map directly onto the EOS input variables (T and P). In other words, any set of reduced properties from the chart is numerically equivalent to the same absolute T and P in the EOS. This approach is purely a mathematical device to permit direct fitting in reduced coordinates - there is no implication that the critical properties of real natural gases are 1 °R and 1 psia. The acentric factor and volume shift were regressed to best match the Standing-Katz Z-factor values over the entire grid of Tr and Pr . The final fit parameters are shown in [Table 11](#).

The resultant single component model can be used as a direct functional replacement (or an initial guess) for iterative Z-factor methods such as DAK, with a concise implementation (see Appendix B). A similar concept has been shared previously ([Burgoyne, Z- Factors for Natural Gas: A simple EOS based approach 2024](#)), but here the method is modified to preserve the default PR omega values for the hydrocarbon pseudocomponent.

Step 2: Pure Inerts. For the four inert components in the model - H₂, N₂, CO₂ and H₂S - we tuned acentric and volume translation factors to match pure component molar densities calculated by the Peng-Robinson EOS against NIST reference data over a broad range of temperatures (50300°F) and pressures (up to 15,000 psi). For CO₂ and H₂S, tuning of OmegaA and OmegaB was also required to deliver satisfactory results over the range investigated. For hydrogen, the critical temperature was modified slightly to conform with that proposed by Nielsen et al (Nielsen M. H. 2023), after confirming that it delivered improved density matches than the default value.

Calibration minimized relative density errors, with all inert gases achieving <2% density error in regions away from the saturation line (vapor, liquid and supercritical areas). In the case of CO₂ and H₂S, since saturation conditions are found within the pressure and temperature regions of interest, tuning was further constrained by incorporating saturation pressure data, ensuring the model reasonably reproduces phase boundaries and transitions for these pure components.

A summary of single component critical properties can be found in Table 11, while maps of relative errors in calculated molar density vs. NIST data for the four pure inert gases clearly demonstrate excellent accuracy achieved by the calibrated parameters in Fig. 3 (H₂), Fig. 4 (N₂), Fig. 5 (CO₂), and Fig. 6 (H₂S).

Step 3: Binary Interaction Parameters. To improve model predictability of mixtures in single phase conditions close to the saturation region, binary interaction parameters (BIPs) were adjusted to predict experimental VLE data of binary and ternary systems summarized in Table 2.

Table 2—Summary of published experimental VLE data used to regress BIP pair relationships

Binary Pairs	H ₂	N ₂	CO ₂	H ₂ S
N ₂	(Streett 1978)			
CO ₂	(Fandino 2015) (Tsankova 2019)	(Fandino 2015) (Westman 2016) (Ottoy 2019)		
H ₂ S		(Besserer 1975)	(A. Chapoy 2013)	
CH ₄	(Freeth 1931) (Benham 1957)	(Han 2012) (Ottoy 2019)	(Hwang 1976) (Legoix 2017) (Petropoulou 2018) (Ottoy 2019)	(Reamer 1951) (Coquelet 2014)

This approach ensures that, once the BIPs have been determined through VLE tuning, the simplified cubic EOS method can confidently be employed in single-phase scenarios adjacent to phase boundaries, benefiting from the enhanced accuracy of mixture phase boundary delineation derived from the calibration. Additionally, a useful "side-effect" of this approach is that if one were to input the EOS equation parameters to a vapor-liquid thermodynamic engine, accurate saturation pressure estimations will be achieved.

To achieve this tuning, the EOS model was implemented in a full-featured PVT modeling tool to determine vapor-liquid equilibrium properties, such as saturation pressures and compositions of liquid and vapor phases.

A temperature-dependent binary interaction parameter relationship was regressed for each binary component pair. The general form of this temperature-dependent BIP is given by:

$$k_{ij} = a + \frac{b}{T_{r_Solvent}} \quad \text{Equation 1}$$

Here, $T_{r_Solvent}$ is the reduced temperature calculated using the critical temperature of the less volatile component ("solvent"). For methane-inert pairs, methane was always treated as the solvent, which effectively generalizes the BIPs to richer hydrocarbon mixtures with higher molecular weights, following the conceptual approach of Valderrama (Valderrama 1990), who found this inverse reduced temperature

scaling was an appropriate representation of temperature dependent k_{ij} behavior for mixtures of hydrogen with a range of n-Alkanes using the single relationship.

For the $H_2:H_2S$ pair, no suitable flash or saturation data could be found in the literature, so a zero value has been assigned.

Coefficients 'a' and 'b' for Equation 1, along with the critical temperature used for calculation of $T_{r_Solvent}$ are outlined in Table 3.

Table 3—Temperature Dependent Binary Interaction Parameter Coefficients

Pair	a	b	Solvent Tc (°R)
Hydrocarbon : CO ₂	-0.14556	0.276572	From Equation 6
Hydrocarbon: H ₂ S	0.16852	-0.12238	From Equation 6
Hydrocarbon : N ₂	-0.108	0.060551	From Equation 6
Hydrocarbon : H ₂	-0.06201	0.042787	From Equation 6
H ₂ S : CO ₂	0.248638	-0.13819	547.416
CO ₂ : N ₂	-0.25	0.11602	547.416
CO ₂ : H ₂	-0.24715	0.16377	547.416
H ₂ S : N ₂	-0.20441	0.234417	672.12
H ₂ S : H ₂	0	0	672.12
N ₂ : H ₂	-0.16625	0.078813	227.16

Step 4: New Hydrocarbon Pseudocomponent Correlation. Tests of several pseudocritical property correlations including Sutton's (gas condensate and associated gas) and PMC showed unacceptable levels of error in the Z-Factor's calculated from our model tuned to S&K Z-Factors. This prompted the development of a new correlation that - when combined with the tuned PR model - better honors both methane and richer gas Z-Factor behavior when evaluated against (a) the Wichert thesis dataset (E. Wichert 1970), with 90 samples and 1,062 data points covering 0.56 - 1.49 SG (predominantly lean) gas, and (b) publicly available data from Geoscience Australia's NOPIMS website (Geoscience Australia 2025) with 61 samples and >1,500 Z-Factor data points covering 0.58 - 2.58 SG gas.

In cases where the source of the gas was not known, discrimination based on C7+/C2 mole fraction ratio was used. When $C7+/C2 > 0.65$, it was deemed to be of associated gas origin, otherwise a gas condensate.

The new correlations for hydrocarbon pseudocomponent critical properties remain functions of hydrocarbon molecular weight (of given origin), and have been regressed to be used in conjunction with the other elements in the proposed model. Regression was performed investigating several forms for Tc and V_c/Z_c , (together from which Pc is calculated) that by construction exactly honors methane's known Tc and Pc. The final version is described in Equation 6 - Equation 9 with parameters in Table 13.

Step 5: Viscosity tuning. The Lohrenz-Bray-Clark (LBC) gas viscosity correlation was selected for viscosity calculations. Three key sets of parameters were tuned: (1) critical volumes (V_{cVis}) for each inert component; (2) slope and intercept of a new correlation for hydrocarbon pseudocomponent V_{cVis} vs hydrocarbon molecular weight; and (3) 4th and 5th coefficients in the LBC viscosity equation.

For rich hydrocarbon gas, we used viscosity data to Lee, Gonzalez and Eakin's (LGE) correlation from 3,000 - 4,000 psia. NIST- reported viscosity data was used for pure methane, CO₂, H₂S, N₂, and H₂ across 14.7-10,000 psia and 50-300°F.

Maps of viscosity relative error as evaluated against NIST for pure components can be found in Fig. 11 to Fig. 15. A summary of critical properties for inerts is presented in Table 11, the relationship of V_{cVis} for hydrocarbons is in Equation 10, and the updated LBC coefficients (with changes to P3 and P4) are in Table 15.

Step 6: Tuning of ideal gas isobaric heat capacity. Ideal gas heat capacities using polynomial expressions based on the functional form originally proposed by Riazi (Riazi 2005) per Equation 2 for inert components

and methane were re-fit to zero-pressure data from NIST for improved accuracy. This re-fitting was necessary to provide sufficient accuracy over the temperature range of interest (450-800 °R or 250-444 K). A new relationship was also created to estimate hydrocarbon pseudocomponent ideal gas Cp values.

Pure component coefficients for ideal gas Cp/R - refitted to NIST data - are presented in Table 4, along with factors to evaluate the ideal gas heat capacity and enthalpy integral in different units in Table 5. Main differences were seen for Methane and CO₂, where RMS improved from 0.70% and 0.65% respectively to < 0.005 %.

$$\frac{C_p^{ig}}{R} = Cp_a + Cp_b T + Cp_c T^2 + Cp_d T^3 + Cp_e T^4 \quad \text{Equation 2}$$

Table 4—Ideal Gas Cp coefficients

Component	Cp _a	Cp _b	Cp _c	Cp _d	Cp _e
CO ₂	2.725473196	0.004103751	1.5602E-05	-4.1932E-08	3.10542E-11
H ₂ S	4.446031265	-0.005296052	2.0533E-05	-2.5899E-08	1.25555E-11
N ₂	3.423811591	0.001007461	-4.5849E-06	8.4252E-09	-4.38083E-12
H ₂	1.421468418	0.018192108	-6.04285E-05	9.08033E-08	-5.18972E-11
CH ₄	5.369051342	-0.014851371	4.86358E-05	-3.7019E-08	1.80641E-12

$$\Delta H_{ig} = \int_{T_{ref}}^T C_p^{ig} \cdot dT = R \cdot \text{Scalar} \cdot \left(\frac{Cp_a(T - T_{ref})}{1} + \frac{Cp_b(T^2 - T_{ref}^2)}{2} + \frac{Cp_c(T^3 - T_{ref}^3)}{3} + \frac{Cp_d(T^4 - T_{ref}^4)}{4} + \frac{Cp_e(T^5 - T_{ref}^5)}{5} \right)$$

Equation 3—Ideal gas delta Enthalpy Equation

Ideal gas enthalpy, being the integral of Cp between two temperature points, can be found in Equation 3, where the units of T for both Equation 2 and Equation 3 remain degrees Kelvin no matter what units of R are used. Additional units and scalar information can be found in Table 5.

Table 5—Heat capacity and enthalpy units and conversions

Cp / Enthalpy Units	Gas Constant R	Scalar
SI: J/(mol·°K) and J/mol	8.312 J/(mol·°K)	1
Field: Btu/(lb-mol·°R) and Btu/lb-mol	1.98588 Btu/(lb-mol·°R)	9/5

To calculate the ideal gas heat capacity for a hydrocarbon pseudocomponent, the Cp coefficient ratios for ethane, propane and butane to methane were examined, and relationships developed to scale each of the methane coefficients from Table 4 as a function of pseudocomponent molecular weight. While (for example) the heat capacity of a natural gas mixture with 1.038 SG may not be identical to ethane (sharing the same MW), these corrections, ed on pure component behaviour, are expected to be sufficiently accurate for natural gas mixtures, and collapse perfectly to methane properties at methane MW. The MW range fitted spans 16.0425 - 58.1222 (0.5538 - 2.0 SG).

$$Cp_{x_Pseudocomponent} = Cp_{x_Methane} (a\Delta MW^2 + b\Delta MW + 1) \quad \text{Equation 4}$$

Where ΔMW is as defined in Equation 9, and other parameters defined in Table 6.

Table 6—Parameters for Cpx_Pseudocomponent Equation

	C _{pa}	C _{pb}	C _{pc}	C _{pd}	C _{pe}
a	7.8570E-04	1.3123E-03	9.8133E-04	1.6463E-03	1.7306E-02
b	-8.1649E-03	5.5485E-03	8.3258E-02	2.0635E-01	2.5551E+00

When evaluating the ideal gas heat capacity of a gas mixture, each component Cpx coefficient (including pseudocomponent) are calculated on a mole fraction weighted average basis per Equation 5, where y_i are the component mole fractions, after which Equation 2 or Equation 3 can be applied to the mixture coefficients.

$$\text{Mixture } C_{p_x} = \sum_{i=1}^5 C_{p_{xi}} \cdot y_i \quad \text{Equation 5}$$

Workflow for Property Calculation:

Using the proposed model for Z-Factor and viscosity involves these steps for a given gas composition and state:

1. Define the mole fraction composition of CO₂, H₂S, N₂, H₂ and hydrocarbon pseudocomponent, along with the hydrocarbon pseudocomponent molecular weight.
2. Calculate the hydrocarbon pseudocomponent's critical T_c, P_c (Equation 6 - Equation 9) and critical volume for viscosity (Equation 10), and retrieve the tuned critical constants for the inerts (Table 11).
3. Calculate temperature dependent BIP pairs at temperature of interest (Equation 1).
4. Use the PR EOS to solve for Z-factor analytically at the given P, T. For most spreadsheet style applications, the largest root can be taken as the gas phase Z-factor.
5. Compute viscosity using the LBC method with the adjusted critical volumes and coefficients from Table 11, Equation 10 and Table 15.

The output is the gas Z-factor and viscosity. This procedure has been implemented in both Python and Excel for verification. A GitHub repository with example code and data is referenced for readers (Burgoyne, 5_Component_PengRobinson_Z-Factor 2025). The model is computationally light and uses standard PR formulation making it suitable for spreadsheet calculations or inclusion as a user-added EOS model in reservoir simulators.

The calculation of thermodynamic properties is somewhat more complex than that needed solely for Z-Factor and viscosity but still has analytic solutions well suited to Python or VBA, using the same inputs as needed for Z-Factors. The steps required for the proposed method include;

1. Retrieve C_{p,ig}/R coefficients Table 4
2. If using a hydrocarbon gas pseudocomponent, calculate the adjusted coefficients C_{pa} - C_{pe} using Equation 4
3. Calculate the sumproduct of each component coefficient with its respective mole fraction using Equation 5. This delivers the 5 coefficients needed to evaluate the ideal gas heat capacity of the mixture
4. Calculate ideal gas heat capacity and enthalpy as needed with Equation 2 and Equation 3, then leveraging the information in Table 5 multiply through by selected value of R and units correction factor if needed.
5. Use established methodology (including temperature dependent BIPs) to calculate departure functions and associated Cp, JT coefficient and enthalpies as needed. A summary of those equations can be found in Chapter 6 of Reid, Prausnitz & Poling (Reid 2000), with detailed implementation available in provided VBA and Python code.

Results

Density Accuracy for Hydrocarbon Gas

Fig. 1 illustrates the relative error map of molar density of Methane for the proposed method vs. NIST data from 30,753 data points over 50 - 300 degrees F and 14.7 - 15,000 psia, showing excellent agreement. Fig. 2 shows the error map for the DAK correlation (using known Methane T_c and P_c) vs. NIST data over the same domain. As evident in the summary in Table 7, the proposed method's accuracy over the range investigated is slightly better than DAK's - especially at high pressures - without any iterative solution scheme required.

Table 7—Summary of Relative Errors from this model and DAK for Methane molar density over 50 - 300 degrees F and 15 - 15,000 psia

Model	Average Relative Error	Average Relative Error at >10,000psia	95% Rel. Errors < than
Proposed model	0.6%	0.4%	1.5%
DAK	0.9%	1.2%	1.6%

For richer hydrocarbons, comparisons against experimental data with real mixtures is more relevant than simple synthetic mixtures as underpinning critical pressure and temperature relationships have embedded assumptions of 'natural' compositional distributions. As hydrocarbon mixtures get richer, models like GERG2008 - with fixed compositional palettes - are increasingly unable to accurately predict behaviour. This is best illustrated in Fig. 18, where predictions for the dataset tested with the proposed approach as well as GERG2008 (with compositions specified to C10) and DAK (with Sutton and Wichert/Aziz corrections) are contrasted for gas condensate and associated gas samples and plotted vs Z-Factor as well as gas specific gravity.

Two main observations are (a) for light hydrocarbon mixtures, the proposed approach is about as good as both GERG2008 and DAK with Sutton & Wichert/Aziz corrections, while (b) as mixtures get increasingly rich the proposed approach appears better than either GERG2008 or DAK. Table 8 summarizes the relative errors of the three models.

Table 8—Z-Factor Relative Error Summary for proposed method and GERG2008

Method	Source	Abs. Avg Rel Error	95% Errors less than
Proposed	Gas Condensate	1.23%	3.02%
	Associated Gas	1.11%	2.75%
GERG2008	Gas Condensate	0.91%	2.84%
	Associated Gas	2.15%	4.78%
DAK + Sutton with Wichert / Aziz	Gas Condensate	1.09%	2.80%
	Associated Gas	4.22%	8.01%

Sour and Inert Gas Mixtures

The accuracy of methods such as the proposed, as well as DAK require determination of mixture critical temperature and pressure. When using pure components these critical properties are known, and a direct comparison can be made. A table summarizing molar density error metrics of the proposed approach, DAK and GERG2008 for pure components - as compared against the 30,753 NIST reference data points over 15 - 15,000 psia and 50 - 300 deg F - can be found in Table 9.

Table 9—Summary of Molar Density Errors for Pure Inerts

Rel Errors: (MDEN-NIST)/NIST	Avg Rel Error				95% Better than			
Model	H ₂	N ₂	CO ₂	H ₂ S	H ₂	N ₂	CO ₂	H ₂ S
Proposed approach	0.6%	0.4%	0.8%	0.7%	1.1%	1.0%	1.3%	1.2%
DAK	3.5%	1.1%	3.5%	5.4%	7.1%	2.6%	7.5%	3.6%
GERG2008	<0.1%	<0.1%	0.1%	0.6%	<0.1%	<0.1%	0.1%	<0.1%

When high concentrations of sour and inert gases are mixed with hydrocarbons, and only gas specific gravity is known, we require critical property estimation methods such as Sutton (with Wichert & Aziz corrections) or PMC. Tests show deficiencies of these traditional methods with such situations with high inerts. To illustrate, a mixture of 5% Methane with 95% CO₂ was simulated with GERG2008, with results contrasted against (a) the proposed approach as well as with DAK using (b) Sutton and Wichert corrections, and then (c) with PMC. This type of mixture would be representative of many CO₂ storage project assumptions when sequestering CO₂ extracted from primary natural gas production.

In such cases, injected temperatures can be reasonably low, so comparisons over 1,000 - 3,000 psia at 40, 100 and 150 degrees F have been made. A hybrid Newton + bisection solver scheme was required for DAK as the standard solution scheme exhibited numerical instability. Even then, numeric problems with DAK persisted for the 40 degF case and initial guesses using the proposed solution were required to successfully resolve the DAK method. Results can be seen in Fig. 9, with a summary of density errors available in Table 10.

Table 10—Summary of Density Errors for 95% CO₂

Errors: (DEN-GERG)/GERG	Avg Rel Error			95% Better than		
Model	40 F	100 F	150 F	40 F	100 F	150 F
Proposed approach	0.7%	2.6%	1.0%	0.7%	1.49%	2.2%
DAK + Sutton & Wichert Aziz	2.3%*	3.8%	5.2%	3.1%*	1.3%	8.2%
DAK + PMC	1.1%*	20.8%	1.2%	16.4%* ¹	39.1%	16.7%

In a hydrogen-methane blend (50% H₂, 50% CH₄) at 40, 100 and 150°F, our model predicts Z-factors that align with GERG2008- calculated values within 2% across 100 - 10,000 psi per Fig. 10. It is not possible to contrast this with traditional correlations since no H₂ support exist for them. These results underline that the new correlation maintains accuracy even at extreme compositions, making it suitable for scenarios like H₂ and CO₂ injection in depleted natural gas reservoirs.

Viscosity Performance

Pure component viscosity errors using the enhanced LBC model have been plotted over the range tested in Fig. 11 (Methane), Fig. 12 (H₂), Fig. 13 (N₂), Fig. 14 (CO₂) and Fig. 15 (H₂S), showing generally <5% error over the entire range for all components.

As a blind test, experimental viscosity data of CH₄ and CO₂ mixtures from Chapoy (Chapoy 2024) - data not included in the regression of the proposed model - was compared against predictions from the proposed model, along with those from the REFPROP ECS and CoolProp Chung viscosity models. They showed similar degrees of error (+/- 10%) over the range of pressures and temperatures that the proposed model has been created for, as evidenced in Fig. 16.

Heat capacity of hydrocarbon mixtures

To test the ideal gas heat capacity relationships proposed in Equation 4, three synthetic mixtures were created as outlined in Appendix C. Plots comparing GERG2008 with proposed ideal heat capacity predictions at

zero pressure over a range of temperatures can be seen in Fig. 17, showing generally lower than 10% difference even for rich gas mixtures

Pressure traverses

Pressure traverses from 15 - 10,000 psia at 60, 150 and 200 degF were made for Density, Viscosity, Enthalpy (relative to standard conditions), isobaric heat capacity (C_p) and Joule-Thomson coefficients of pure components using the proposed model and NIST data, with results available in Fig. 19 to Fig. 23.

Blind test for oilfield gases

A blind test was conducted against the dataset published from (Mogensen, K 2021). The 1,469 measured Z-Factor values were contrasted with predictions from this model with an average absolute relative error of 1.96% as can be seen in Fig. 24, contrasting well versus the alternative method evaluated by Mogensen.

Tabulated Parameter Set

The final tuned (constant) parameters for the EOS model are summarized for reference in Table 11. Additional functions and data required for calculation of pseudocomponent T_c , P_c and V_{cVis} can be found in Equation 6 through Equation 10, and Table 13 through Table 15.

Table 11—Summary of Critical properties

Component	MW	T_c (°R)	P_c (psia)	ACF	VTRAN	OmegaA	OmegaB	VcVis (ft ³ /lbmol)
CO ₂	44.01	547.416	1069.51	0.12253	-0.27607	0.427671	0.0696397	1.46352
H ₂ S	34.082	672.12	1299.97	0.04909	-0.22901	0.436725	0.0724345	1.46808
N ₂	28.014	227.16	492.84	0.03700	-0.21066	0.457236	0.0777961	1.35526
H ₂	2.016	47.43	187.53	-0.21700	-0.3627	0.457236	0.0777961	0.68473
Pseudocomponent		Equation 6 - Equation 9		-0.03899	-0.19076	0.457236	0.0777961	Equation 10

Table 12—Pseudocomponent Critical relationships

$$T_c(^{\circ}\text{R}) = \frac{a \cdot \Delta MW}{b + \Delta MW} + c \quad \text{Equation 6}$$

$$P_c (\text{psia}) = \frac{R \cdot T_c}{\left(\frac{V_c}{Z_c}\right)} \quad \text{Equation 7}$$

$$\left(\frac{V_c}{Z_c}\right) (\text{ft}^3/\text{lbmol}) = \text{Slope}_{V_c} \cdot \Delta MW + \left(\frac{V_c}{Z_c}\right)_{CH_4} \quad \text{Equation 8}$$

$$\Delta MW = MW_{Hydrocarbon} - MW_{CH_4} \quad \text{Equation 9}$$

Table 13—Pseudocomponent T_c & P_c relationships parameters

Hydrocarbon Type	a	b	c	Slope _{Vc}
Gas Condensate	1098.1095	101.5292	T_{cCH_4}	0.170931
Associated Gas	2695.1477	274.3417	T_{cCH_4}	0.177498

Table 14—Methane Critical Properties

T_{cCH_4} (°R)	P_{cCH_4} (psia)	$\left(\frac{V_c}{Z_c}\right)_{CH_4}$ (ft ³ /lbmol)	MW_{CH_4} (lb _m /lbmol)
343.008	667.029	5.518526	16.0425

$$V_{cVIS}Pseudocomponent (\text{ft}^3/\text{lbmol}) = 0.0576710 \cdot \Delta MW + 1.44383 \quad \text{Equation 10}$$

Table 15—Modified LBC Coefficients (P3 and P4)

P0	P1	P2	P3	P4	F0
0.1023	0.023364	0.058533	-0.0392852	0.00926279	0.1

In Fig. 7, the new Tc & Pc correlations for the hydrocarbon pseudocomponent is compared against Sutton's Gas Condensate and Associated gas methods (R. Sutton 2007), showing that our proposed relationships for Tc fall between Sutton's trends - tending toward gas condensate at lower MW and associated gas at higher MW - while the Pc relationships follow closer to Sutton's associated gas trend. Importantly, at methane MW, our proposed relationship delivers exact solutions for Methane unlike the Sutton gas condensate relationship. It is interesting to note that the data regressed for this work points to little difference between associated gas or gas condensate critical parameters at SG's less than 1.0.

Also of interest is a comparison of the (Vc/Zc) relationships (Fig. 8), which shows distinct curvature up and down in the implied Sutton relationships ((Vc/Zc) = R • Tc / Pc).

Discussion

Broader Applicability and Reliability

The new EOS-based correlation demonstrates broad applicability - from methane-dominant natural gas to 100% CO₂ or 100% H₂ cases - with low error over wide ranges of application. This is an expansion over traditional methods that were confined to a narrow band of compositions. For engineers, this means one can confidently use a single model for scoping, whether dealing with natural gas production (mostly hydrocarbons) or later injection for storage (rich in CO₂, H₂ or N₂). The method essentially provides a "drop-in" replacement for DAK / Hall-Yarborough Z-Factor and associated approaches for critical properties, viscosity and thermal properties. It is easy to implement (no iterative solves, formulas suitable for coding or Excel) and eliminates the patchwork of supplemental correlations previously needed for special gases and viscosity.

Implications for CCUS and Gas Storage

In carbon capture and storage (CCS/CCUS) projects, pure CO₂ or CO₂-rich gases are injected into reservoirs. Accurate prediction of CO₂ compressibility and viscosity is crucial for estimating injection rates, storage capacity, and pressure buildup. Our model directly addresses this by calibrating on high-quality CO₂ data. Similarly, the rise of studies on underground hydrogen storage in depleted gas fields means engineers must have ways to model H₂-natural gas mixtures and even pure hydrogen in reservoir and pipeline calculations. Legacy Z-factor correlations do not support H₂, whereas our EOS approach naturally accommodates H₂. Even inert gas injection or storage (such as N₂ used as a cushion gas in natural gas storage facilities, or flue gas injection containing N₂) can be handled without special treatment. In summary, the new model is well-timed to support these emerging applications, providing a ready tool for scenarios that previously lay outside everyday engineering experience.

Practical Deployment

A key advantage is that the model can be easily deployed in standard tools. For example, an Excel workbook can implement the analytic cubic solution to yield gas Z-factor with a modest amount of code. This gives engineers a quick-look capability for "what-if" scenarios (e.g., varying H₂ injection percentages) without needing specialized PVT software. Moreover, the method can be implemented in reservoir simulators via user-defined PR EOS critical properties, unlike GERG-2008 which would often require complex coding or built-in support. The simplicity does not come at the cost of physics because it's based on an industry standard EOS so the model behaves reasonably outside the calibration range and conserves correct limiting

behavior (e.g., ideal gas limit at low pressure). This is an improvement over some correlations that might give unphysical results when extrapolated.

Limitations and Future Work

It should be noted that the implementation of this model assumes single-phase conditions. If a gas mixture is near its dewpoint or if liquid dropout occurs, the method will need to be coupled with a phase-split calculation to identify the gas composition - automatically supported with most commercial EOS implementations. Temperature-dependent BIPs helps push the accuracy of predicting where phase separation occurs for inert and methane pairs and can reasonably predict not only single phase vapor but also single-phase liquid properties. The current formulation covers only the five components (C_i pseudo, CO_2 , H_2S , N_2 , H_2), in cases where other components are significant (e.g., helium in underground gas storage, or heavier hydrocarbons in rich gas where accurate dew point is important), extension or additional tuning would be required. However, the framework is extensible: new components can be added with the same tuning philosophy (calibrate critical properties and interaction parameters to data).

Future work could explore adding helium or oxygen for applications like flue gas or validating the model for high-heavies content (retrograde gases) by adding a second pseudo-component if needed. A second area of potential development is to incorporate temperature dependency in volume translation (rather than constant shift) to see if it further improves density matches, though our results show the current approach is already reasonably accurate while avoiding complexities of implementing temperature dependence in commercial simulation tools (not commonly supported). A third area of development would be to refine the critical property calculation method to permit incorporation of parameters such as Watson characterization factor to better describe the PNA character of heavier hydrocarbons and permit more accurate T_c and P_c predictions.

Conclusions

Unified EOS-Based Correlation

We have developed a universal Peng-Robinson EOS-based correlation that provides accurate gas compressibility factors, viscosities and thermal properties for mixtures spanning from pure hydrocarbons to 100% inert components. It consolidates what used to require multiple correlations and adjustments into one coherent model, without sacrificing ease of use.

It can serve as a practical replacement for commonly used Z-factor correlations like DAK and Hall-Yarborough, while also incorporating what would otherwise need separate viscosity and sour gas calculations. The model's parameters and formulas are provided openly, and implementation does not require specialized software. Engineers and simulator vendors can adopt this method to improve the fidelity of their predictions without adding complexity.

Accuracy and Efficiency

The new method achieves accuracy on par with or better than traditional Z-factor correlations in their domain and significantly extends the reliable range to high CO_2 , H_2S , N_2 and H_2 containing gases. By analytically solving the PR EOS for the largest root, the approach can avoid iterative calculations, making it as computationally efficient as empirical formulas - suitable for real-time calculations or spreadsheet implementation. The incorporation of volume translation and targeted tuning yields low errors in density (Z) and viscosity when benchmarked against trusted data sources.

Broad Applicability

This correlation is immediately applicable to modern challenges: Carbon storage operations can use it to model pure CO_2 injection behavior more accurately than legacy methods; hydrogen blending and storage

projects gain a validated tool for H₂-methane mixtures; and natural gas systems with high N₂ (e.g., N₂ injection for pressure maintenance or gas cycling) can be handled within the same framework.

Nomenclature

Roman Symbol	Description	Units
ACF	Acentric factor	-
C _p	Isobaric (constant pressure) heat capacity	J/(mohK), Btu/(lbmoh°R)
C _{p_ig}	Ideal gas isobaric heat capacity	J/(mohK), Btu/(lbmoh°R)
CP _a - CP _e	Ideal gas C _p polynomial coefficients (Eq. 2)	-
CP _x	Any of the ideal gas C _p polynomial coefficients a - e	-
dH	Enthalpy difference	J/mol/Btu/lbmol
k, k _{ij}	Binary interaction parameter (between components i, j)	-
MW	Molecular weight (component or mixture)	lbm/lbmol, g/mol
Omega _A , Q _A	Peng-Robinson EOS attraction parameter constant	-
Omega _B , Q _B	Peng-Robinson EOS covolume parameter constant	-
P	Pressure	psia, atm, MPa
P ₀ - P ₄	LBC viscosity correlation coefficients	-
P _c	Critical pressure	psia, atm, MPa
P _r	Reduced pressure (P/P _c)	-
R	Universal gas constant	psift ³ /(lbmoh°R), J/(mohK), Btu/(lbmoh°R)
SG	Gas specific gravity (MW _{gas} / MW _{air})	-
T	Temperature	°R, °K, °F, °C
T _c	Critical temperature	°R, °K
T _r	Reduced temperature (T/T _c)	-
v	Molar volume	ft ³ /lbmol, m ³ /kmol
V _c	Critical molar volume	ft ³ /lbmol, m ³ /kmol
V _{cVis}	Critical molar volume for viscosity calculation	f ³ /lbmol, m ³ /kmol
VTRAN	Volume translation factor for PR EOS	-
W	Weighting factor (regression/objective function)	-
y _i	Mole fraction in gas phase (of component i)	-
Z	Gas compressibility factor (Z-Factor)	-
Z _c	Critical compressibility factor	-

Greek Letter	Description	Units
μ (mu)	Dynamic viscosity	cP, uPa·s
ρ (rho)	Density	lbm/ft ³ , kg/m ³
γ	Gas specific gravity (MW _{gas} / MW _{air})	-
ω	Acentric factor (EOS parameter)	-

Subscript / Superscript	Description
c	Critical property
EOS	Property calculated directly from cubic EOS
i	Property of component i
ij	Property / interaction between components i and j
pc	Pseudocritical property (mixture)
r	Reduced value (property / critical value)
Vis	Viscosity calculation

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Appendix A

Figures

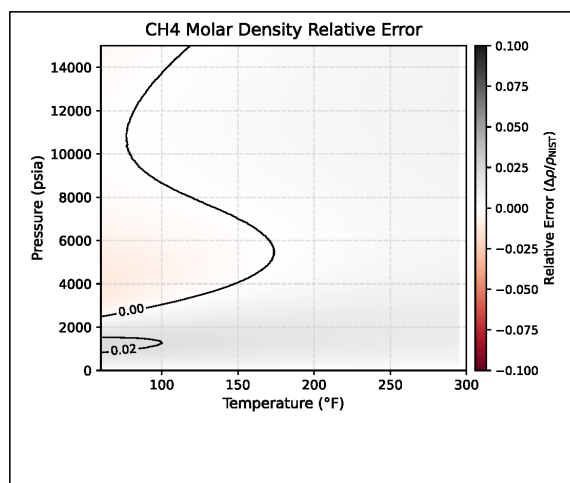


Figure 1—Relative density error with proposed approach for CH₄

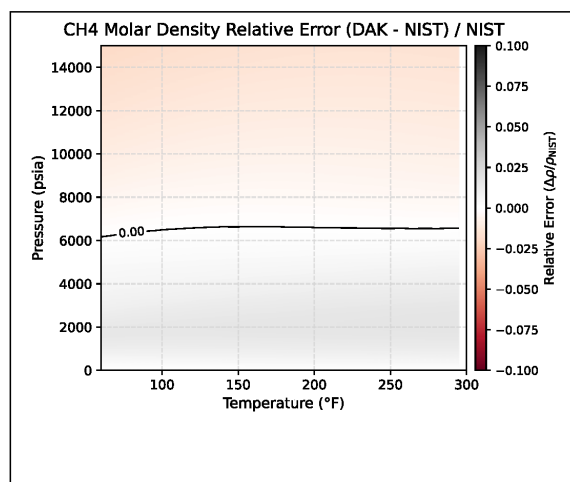


Figure 2—Relative density error with DAK for CH₄

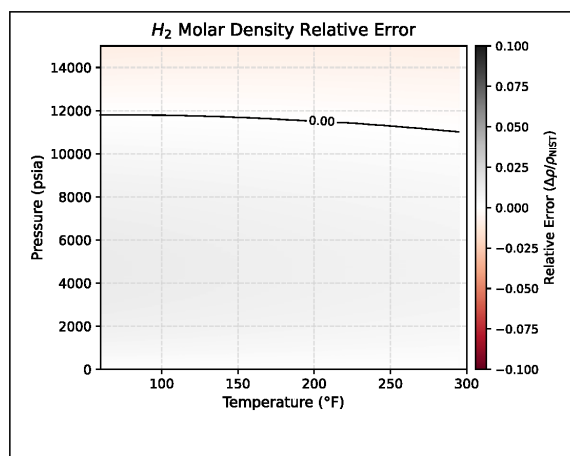
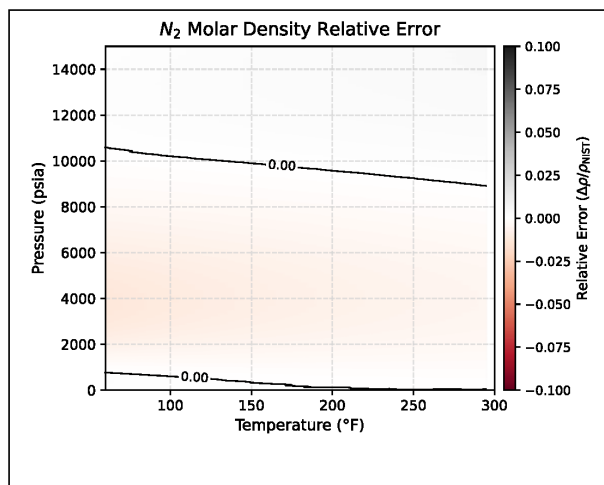
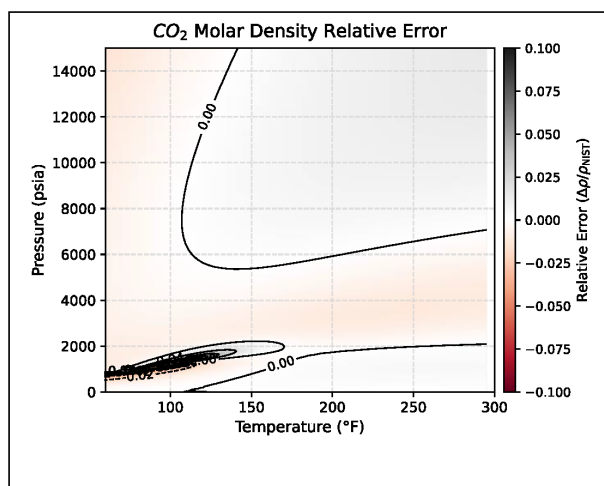
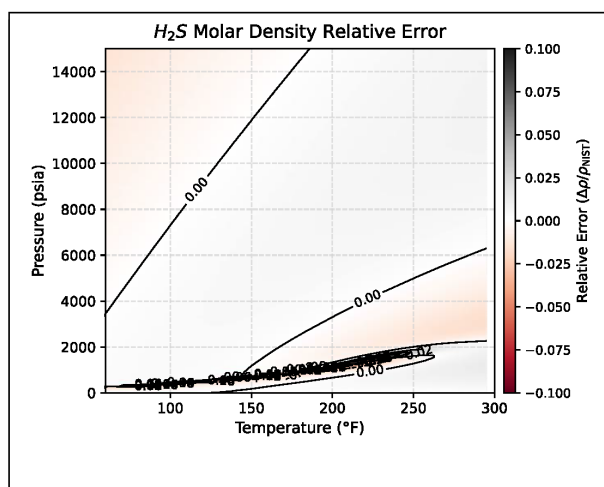


Figure 3—Relative density error with proposed approach for H₂

Figure 4—Relative density error with proposed approach for N₂Figure 5—Relative density error with proposed approach for CO₂Figure 6—Relative density error with proposed approach for H₂S

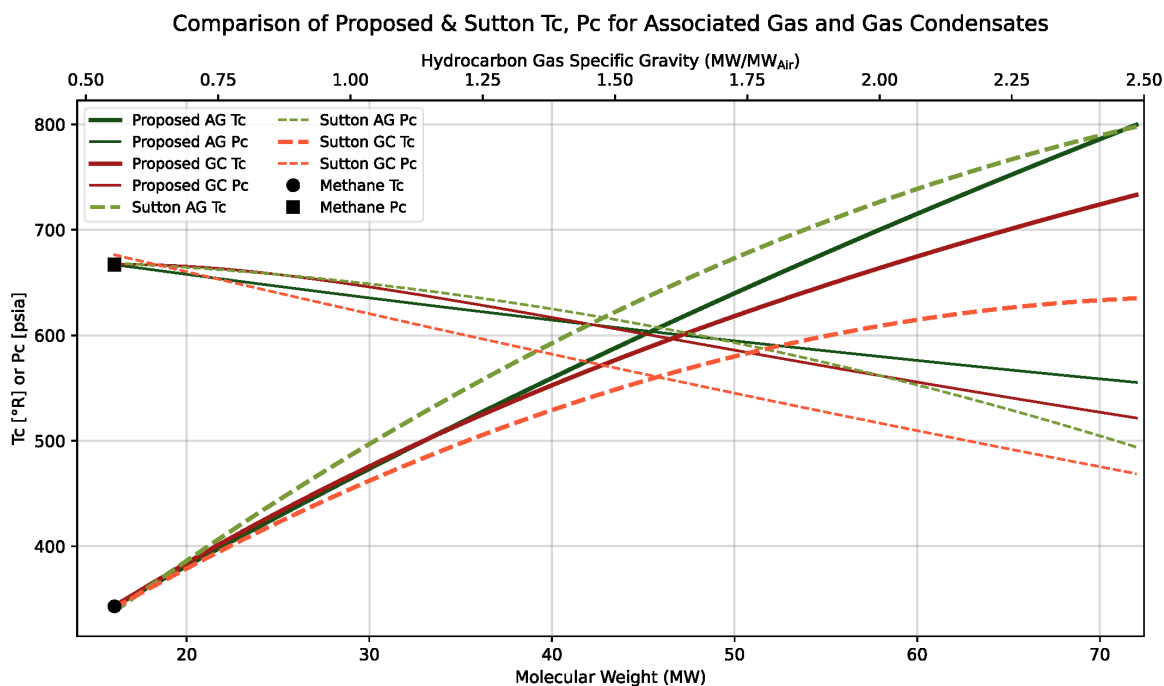


Figure 7—Comparison of Tc and Pc relationships

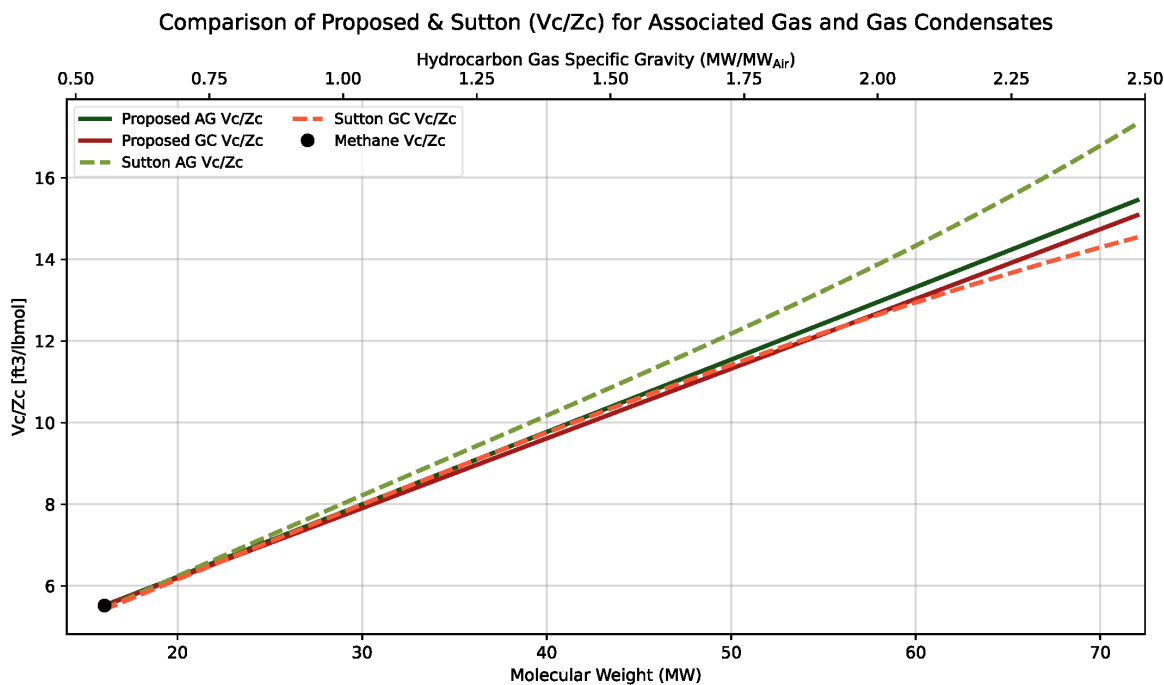


Figure 8—Comparison of proposed and implied Vc/Zc relationships

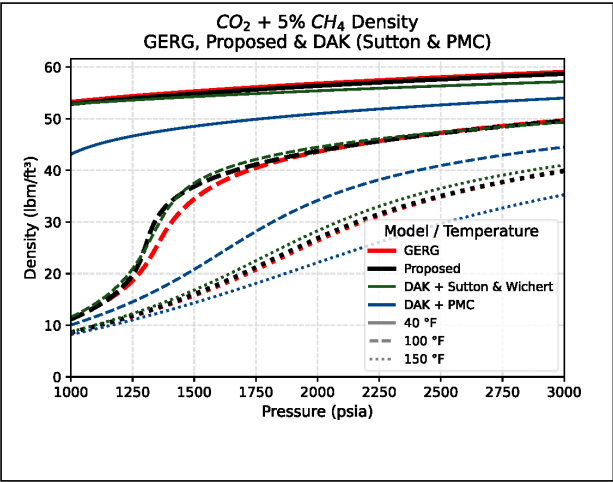


Figure 9—Comparison of 95% CO₂ Density with 4 Methods

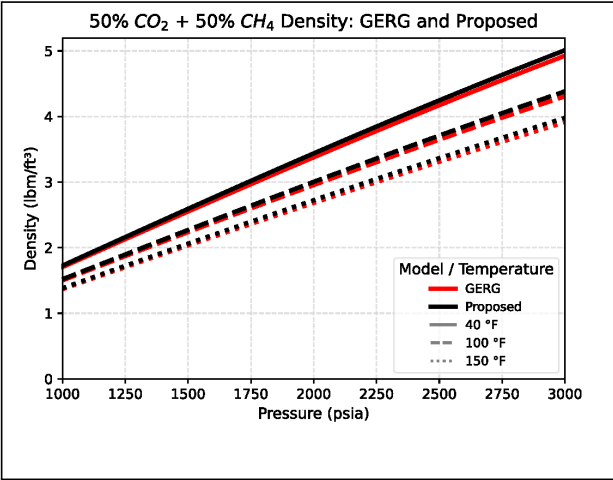


Figure 10—Contrasting Calculated Density of 50:50 H₂ & CH₄

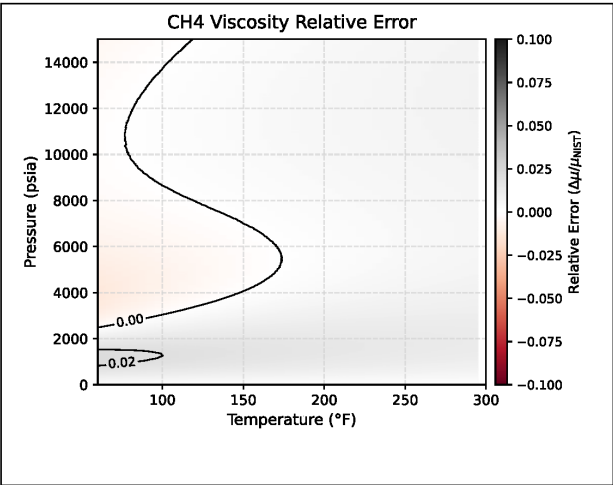


Figure 11—Relative viscosity error with proposed approach for CH₄

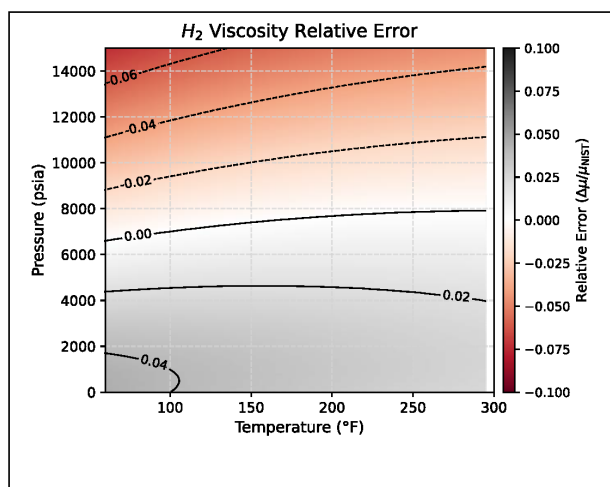


Figure 12—Relative viscosity error with proposed approach for H_2

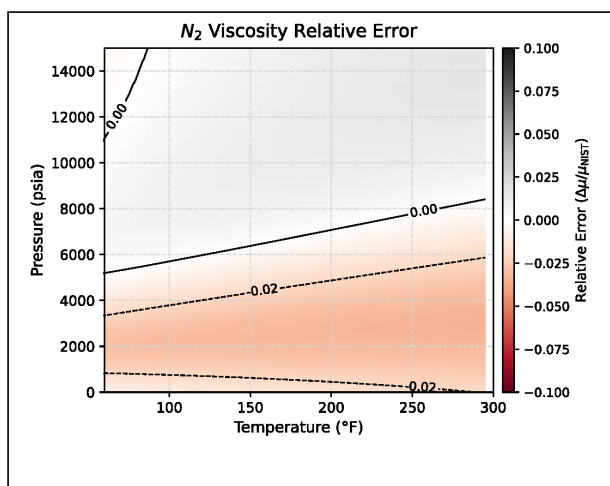


Figure 13—Relative viscosity error with proposed approach for N_2

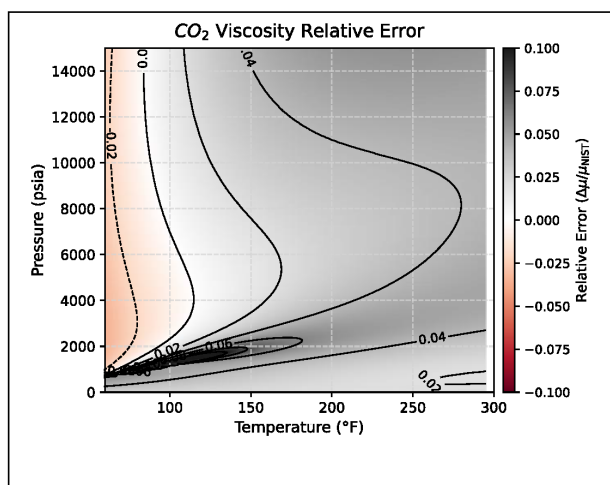


Figure 14—Relative viscosity error with proposed approach for CO_2

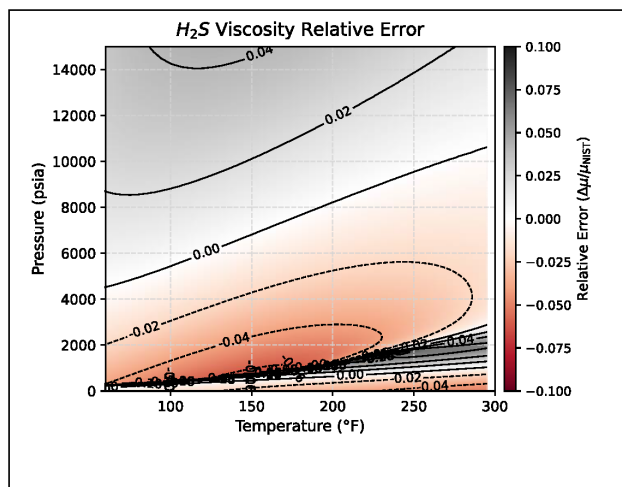
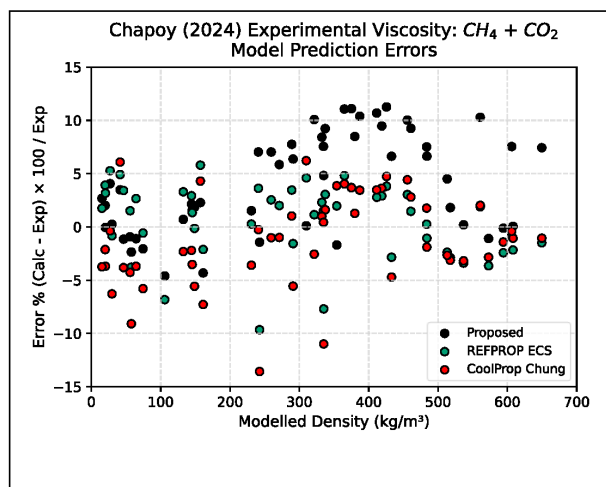
Figure 15—Relative viscosity error with proposed approach for H₂S

Figure 16—Blind test Methane + Carbon Dioxide mixture viscosities

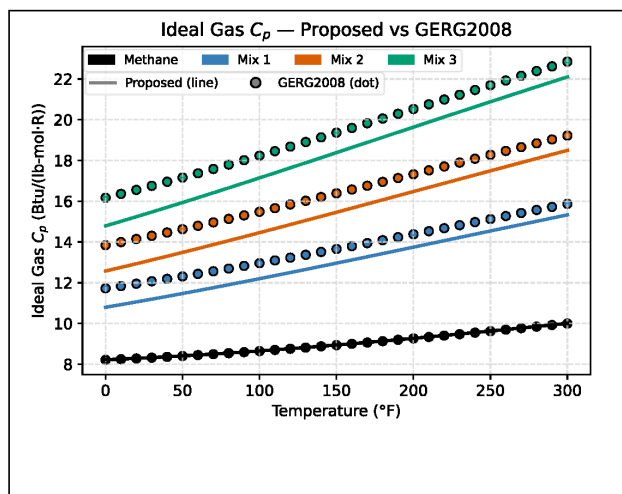


Figure 17—Comparison of ideal gas heat capacity with proposed and GERG2008 models

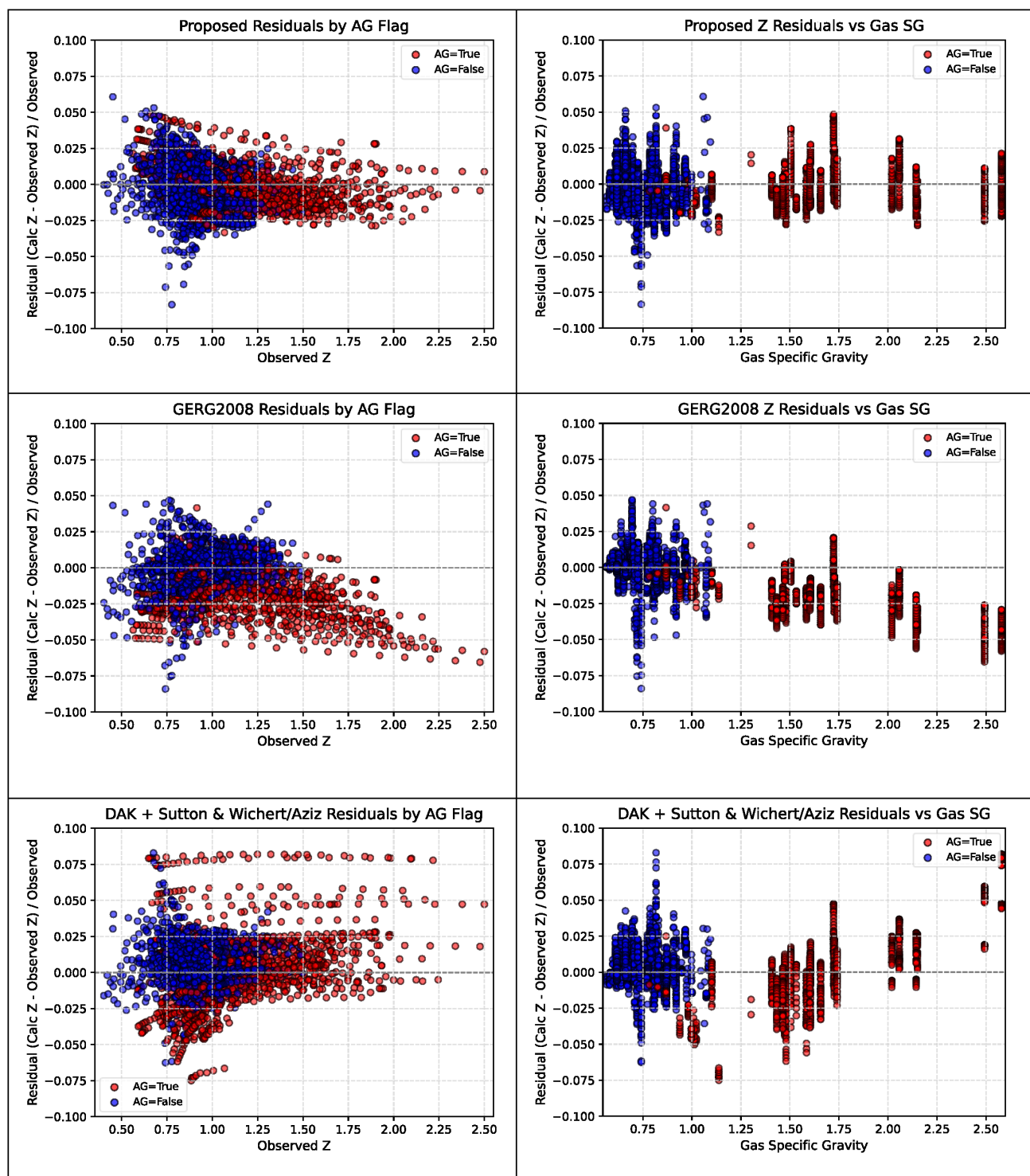


Figure 18—Proposed, GERG2008 & DAK Residuals, split by Associated Gas (AG) origin. Plotted vs Observed Z-Factor and Raw Gas SG

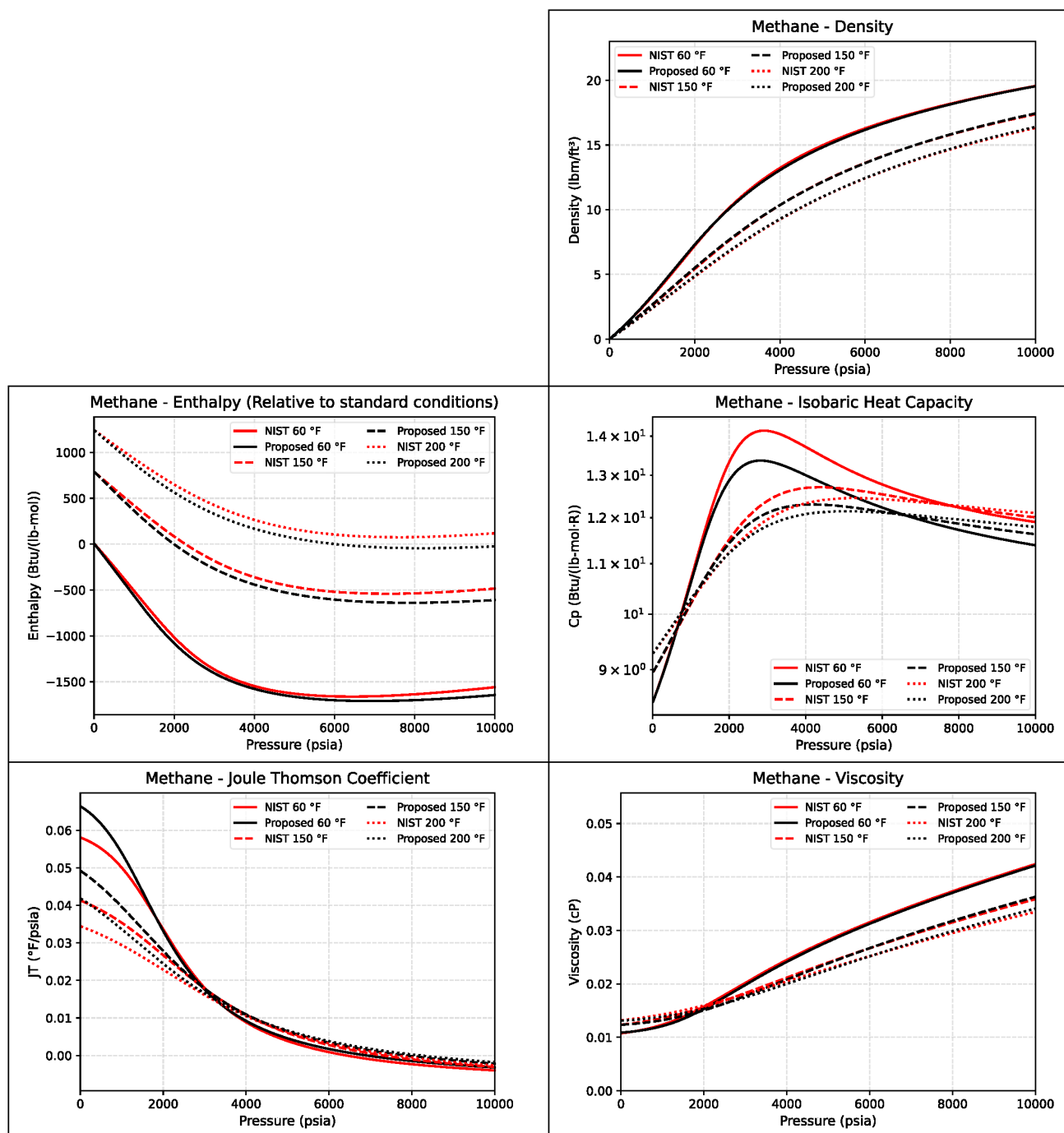


Figure 19—Methane Pressure Traverse Results

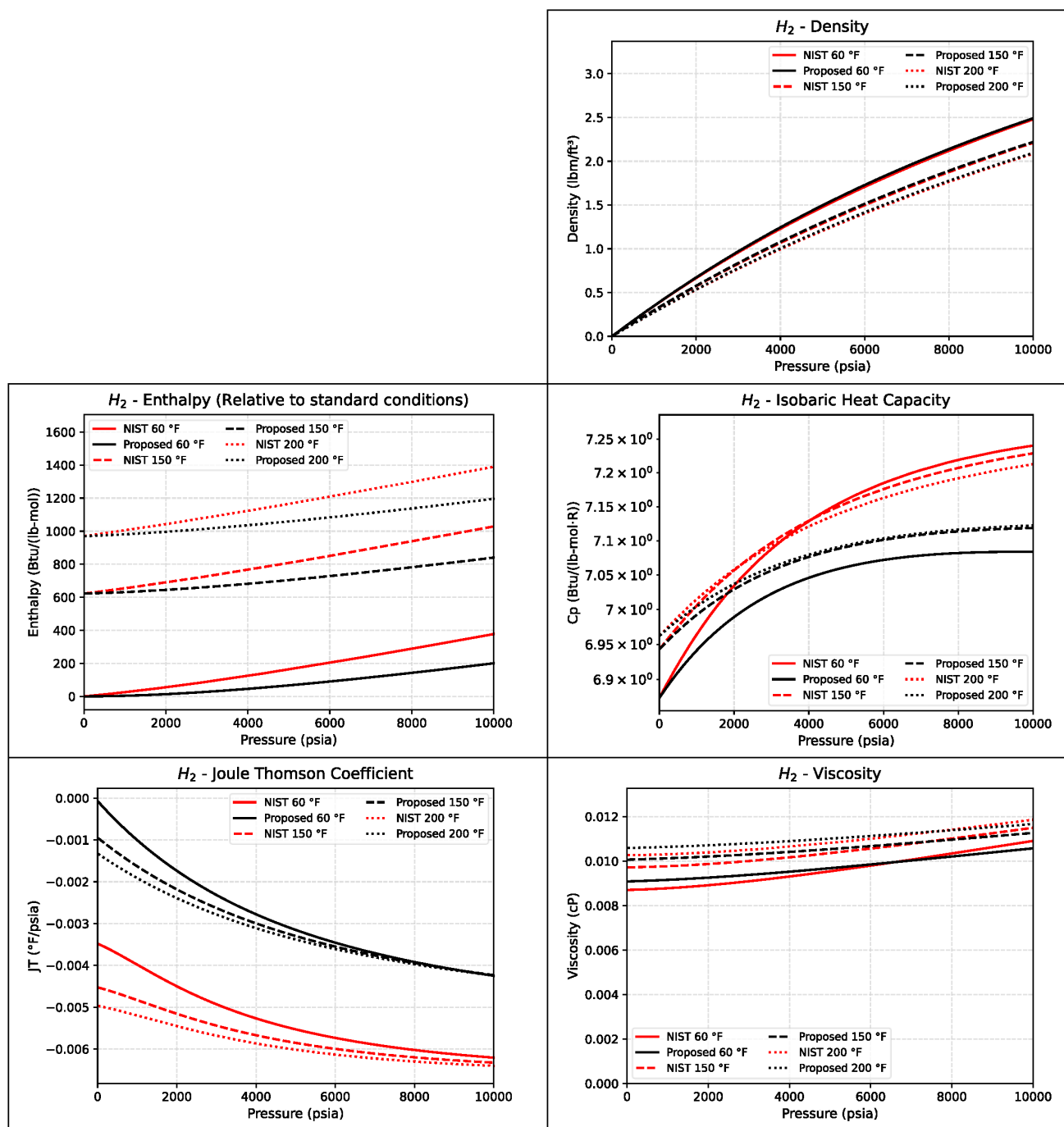
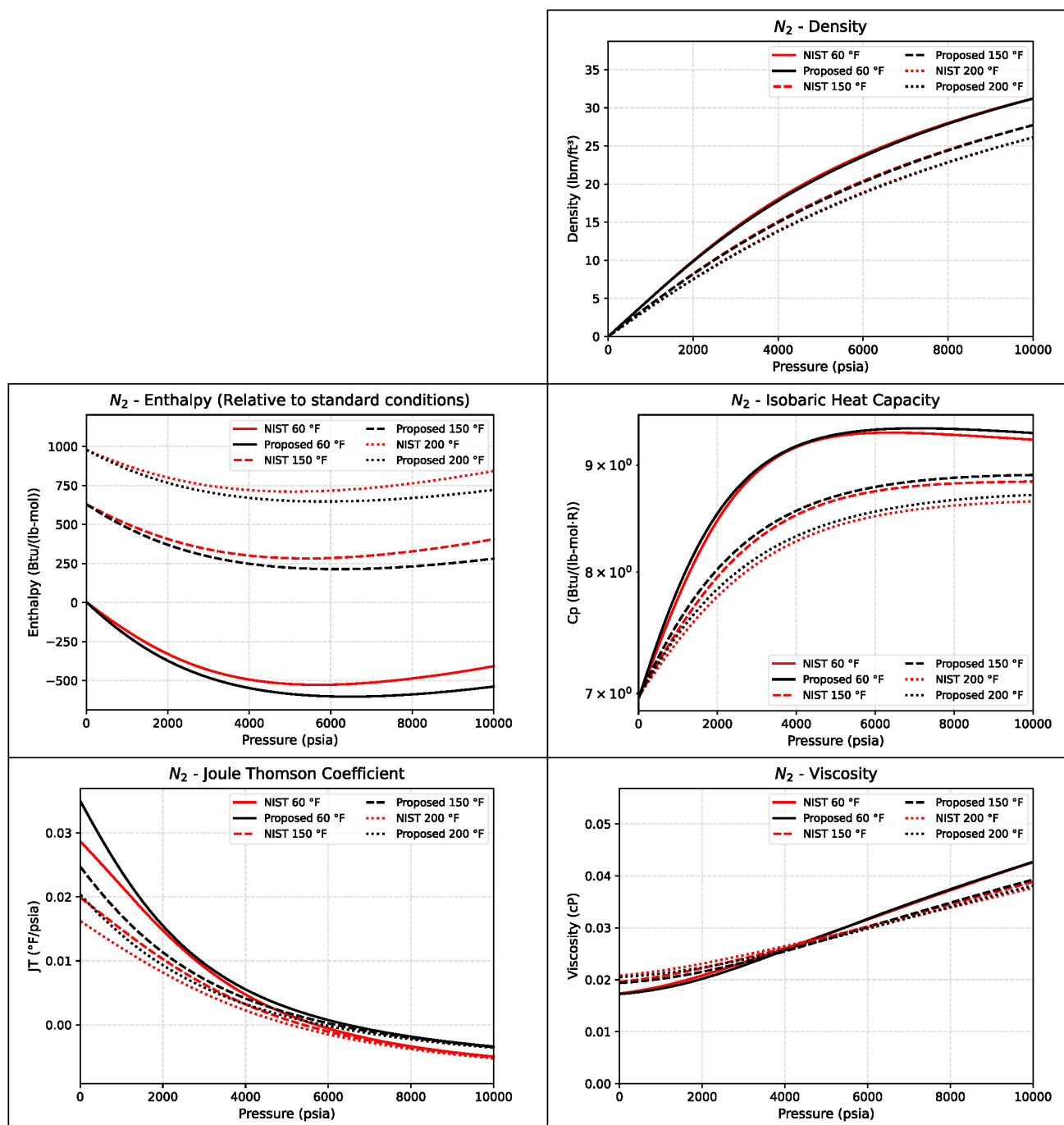
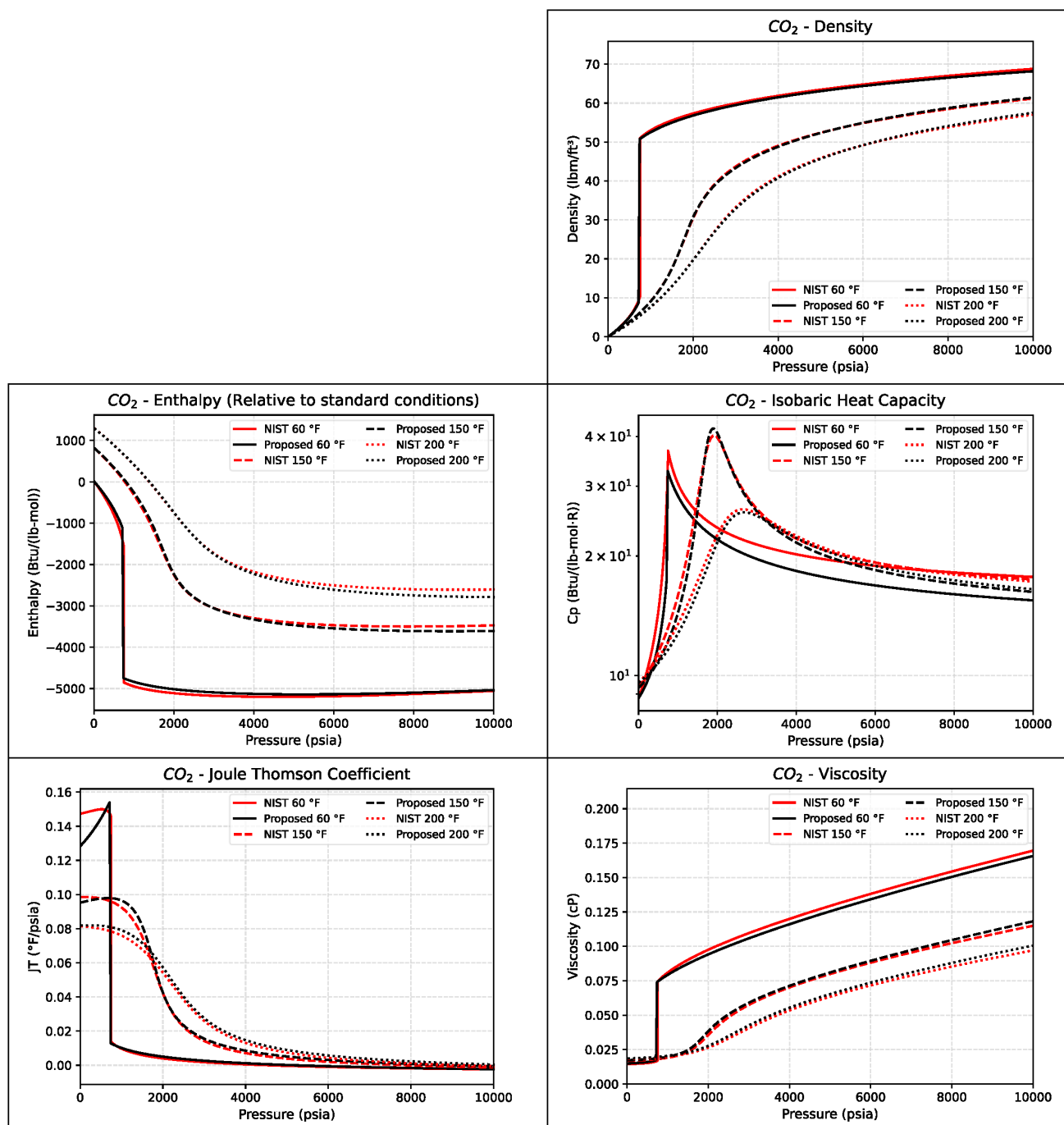
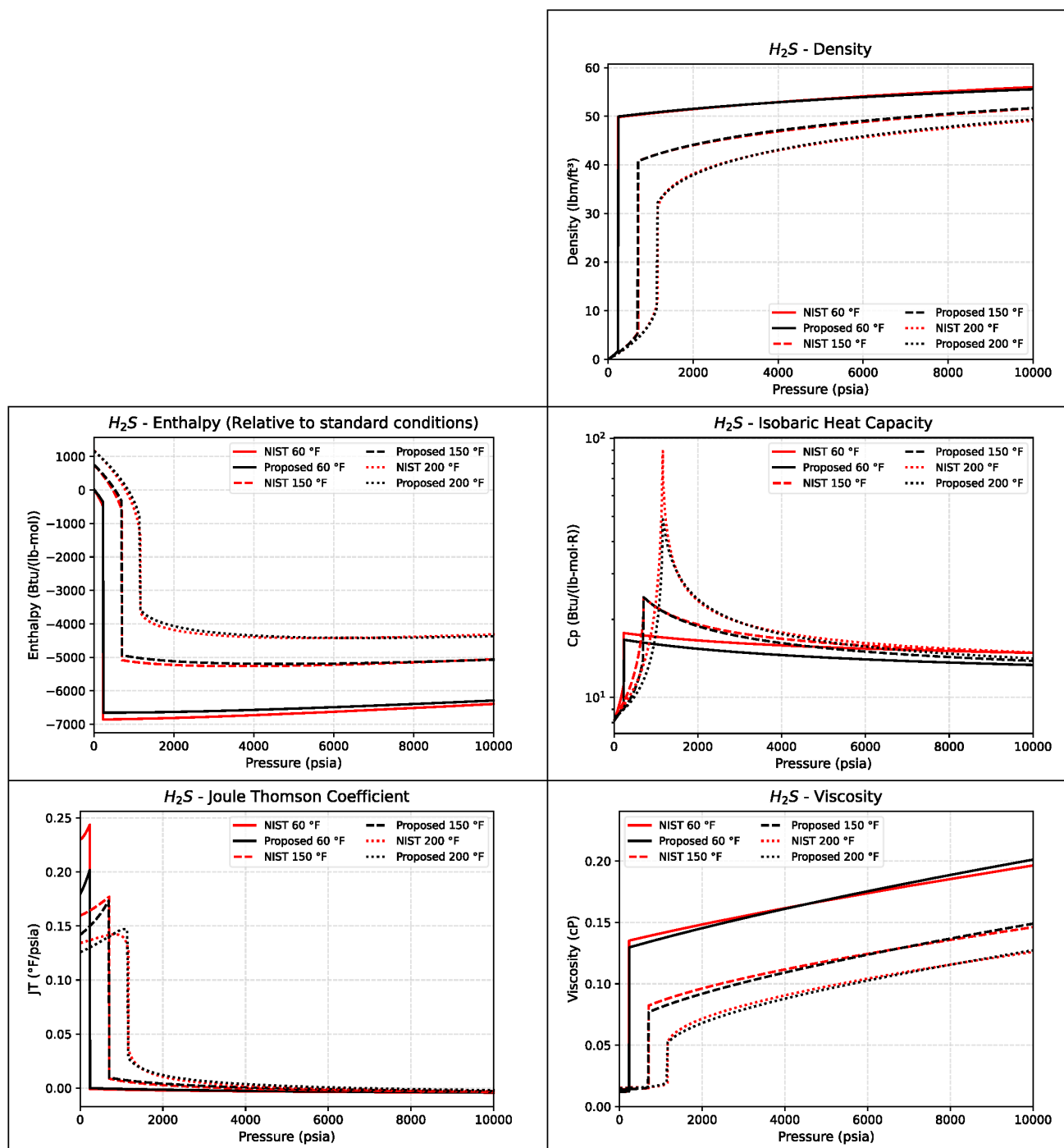


Figure 20—H2 Pressure Traverse Results

Figure 21— N_2 Pressure Traverse Results

Figure 22—CO₂ Pressure Traverse Results

Figure 23— H_2S Pressure Traverse Results

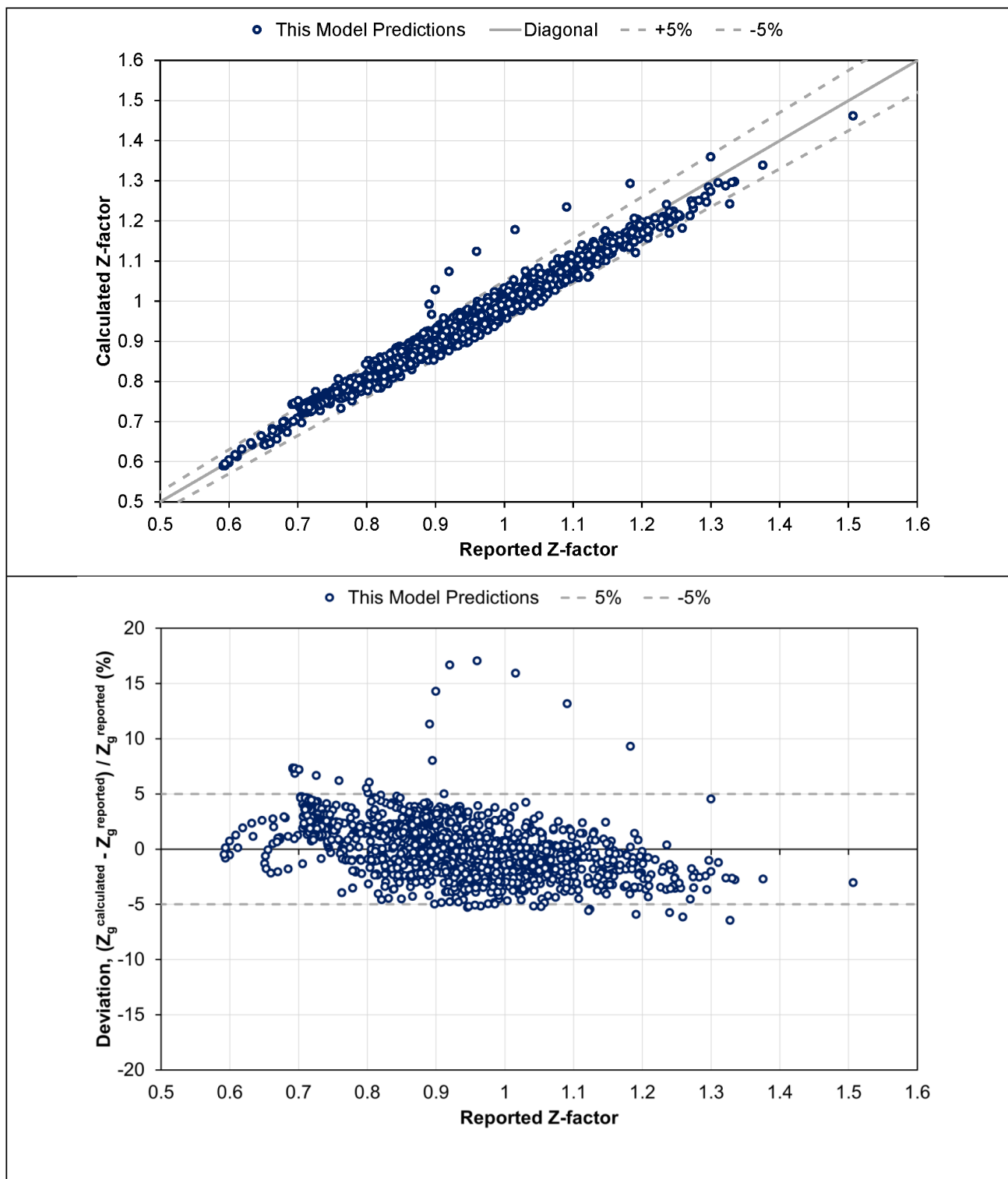


Figure 24—Blind Test Cross Plot and Deviation Plot with SPE207979 Data

Appendix B

Single Component Z-Factor Algorithm Python code

As a direct replacement - or even to provide initial guesses - for iterative solutions of natural hydrocarbon gas mixture Z-Factors, the single component version of this Peng Robinson method can be used with the Acentric Factor and Volume Translation Factor from Step 1: Pure Hydrocarbon gases. A minimal working Python code is presented here as an example that takes reduced temperature and pressure as input and returns the Z-Factor conforming with Standing & Katz data.

Similarly, pure component Z-Factors can be evaluated for H₂, N₂, CO₂ or H₂S by changing the values of *w* (ACF), *s* (VTRAN), *OmegaA* and *OmegaB* in the below code to those corresponding to each pure component from [Table 11](#). Effectively, this is a four parameter analytically evaluated Z-Factor correlation with the robustness of an EOS.

By assuming the largest positive real root will always be the solution, we eliminate inner and outer loops to iteratively solve the problem, simplifying the solution scheme by providing an analytic closed form (when using an analytic cubic root finding method). A numerical root solving method has been used here for simplicity, leveraging available Python libraries.

```
import numpy as np
def max_root(coeffs):
    """Return the largest real root of a cubic (for PR EOS)."""
    roots = np.roots(coeffs)
    real_roots = [r.real for r in roots if abs(r.imag) < 1e-8]
    return max(real_roots)

def pr_z(tr, pr):
    """Single-component Peng-Robinson Z-factor with volume shift."""
    w, s, OmegaA, OmegaB = -0.03893894, -0.190762, 0.457236, 0.0777961
    m = 0.37464 + 1.54226 * w - 0.26992 * w**2
    alpha = (1 + m * (1 - np.sqrt(tr)))**2
    A = OmegaA * alpha * pr / tr**2
    B = OmegaB * pr / tr
    coeffs = [1, -(1 - B), A - 3*B**2 - 2*B, -(A*B - B**2 - B**3)]
    return max_root(coeffs) - s * B # Return largest real root with volume shift applied
```

Appendix C

Synthetic hydrocarbon gas mixtures tested

Component	MW	Mix 1	Mix 2	Mix 3
C1	16.042	0.7082	0.6305	0.5719
C2	30.069	0.0981	0.097	0.0876
C3	44.096	0.0651	0.0715	0.0697
iC4	58.122	0.0216	0.0264	0.0277
nC4	58.122	0.0216	0.0264	0.0277
iC5	72.149	0.0143	0.0195	0.0221
nC5	72.149	0.0143	0.0195	0.0221
C6	86.175	0.019	0.0287	0.0351
C7	100.202	0.0126	0.0212	0.0279
C8	114.229	0.0084	0.0156	0.0222
C9	128.255	0.0056	0.0115	0.0177
C10	142.282	0.011	0.0323	0.0682
Mixture MW		27.913	34.512	41.627
SG rel. Air		0.964	1.191	1.437